

Evaluation of the amplitude for different AC signals

Students of the academic year 2024/2025

16/03/2025



UNIVERSITÀ
DELLA
CALABRIA

DIPARTIMENTO DI INGEGNERIA
INFORMATICA, MODELLISTICA,
ELETTRONICA E SISTEMISTICA

DIMES

**MEASUREMENTS FOR TELECOMMUNICATION
SYSTEMS** by Prof. Francesco Lamonaca
A.A. 2024/2025

This work is part of the project:

EXTENDABLE

NETWORK OF **EXTENDED REALITY-ENABLED**
LABORATORIES FOR REMOTE PRACTICAL TRAINING

Web-site: <https://extendable.dieti.unina.it/>



Finanziato
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Ministero
dell'Università
e della Ricerca



Italiadomani
PIANO NAZIONALE
DI RIPRESA E RESILIENZA

**The Research Team involving Units from:
University of Calabria, Naples Federico II, and Sannio
is proposing the creation of a network of laboratories**



Process

Goals

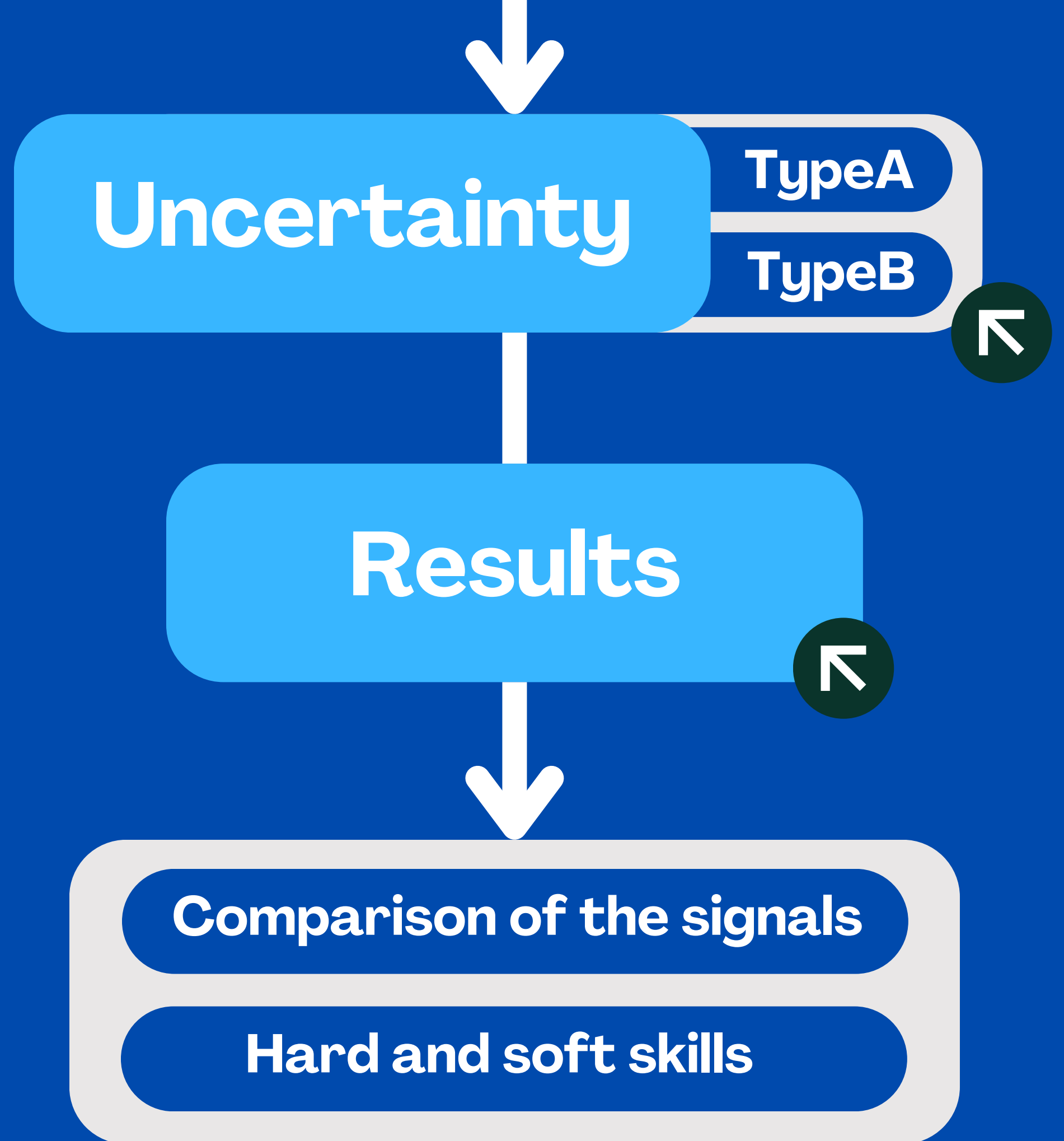
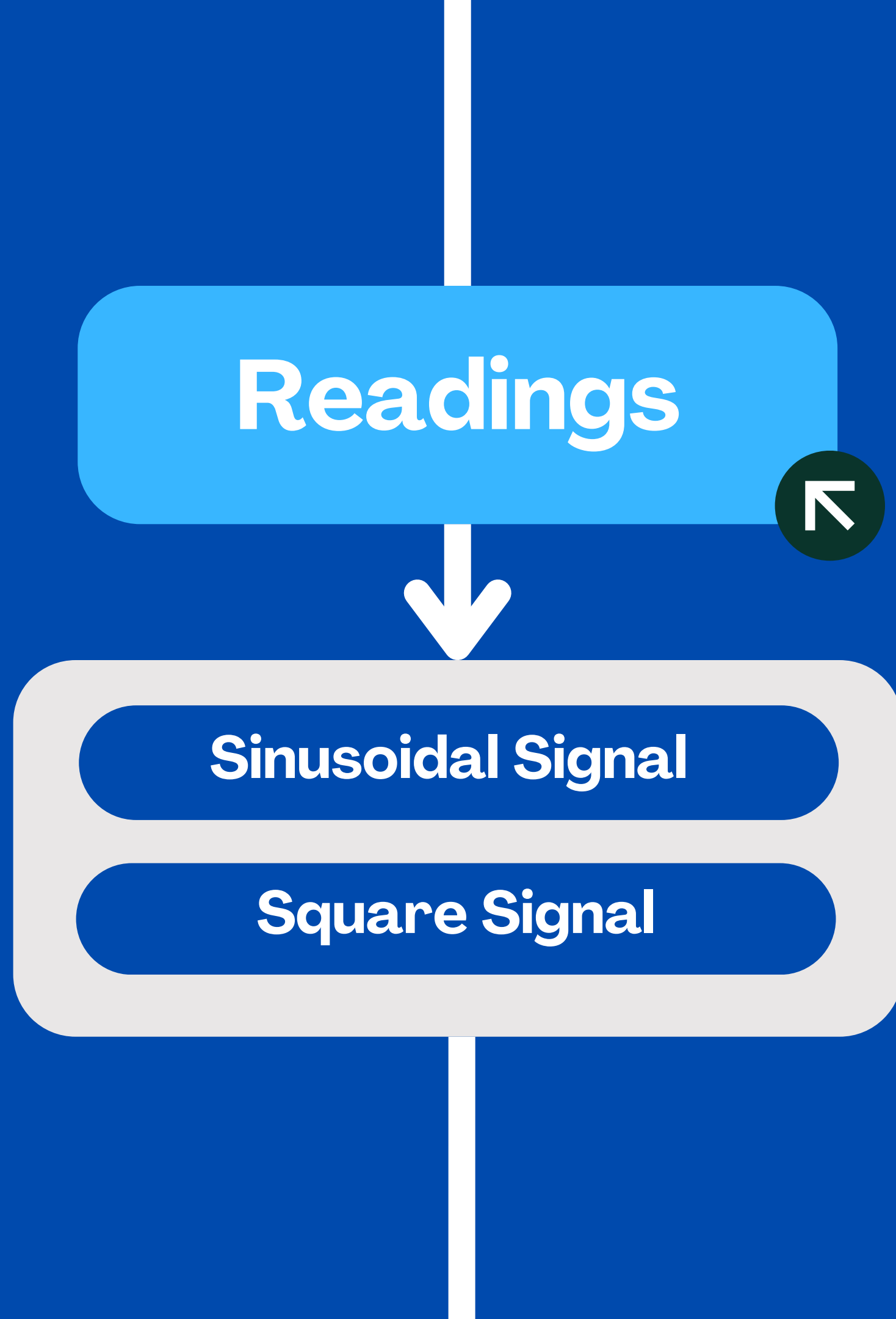
Instruments

Questions?

What are the instruments?

How to link them?





Goals

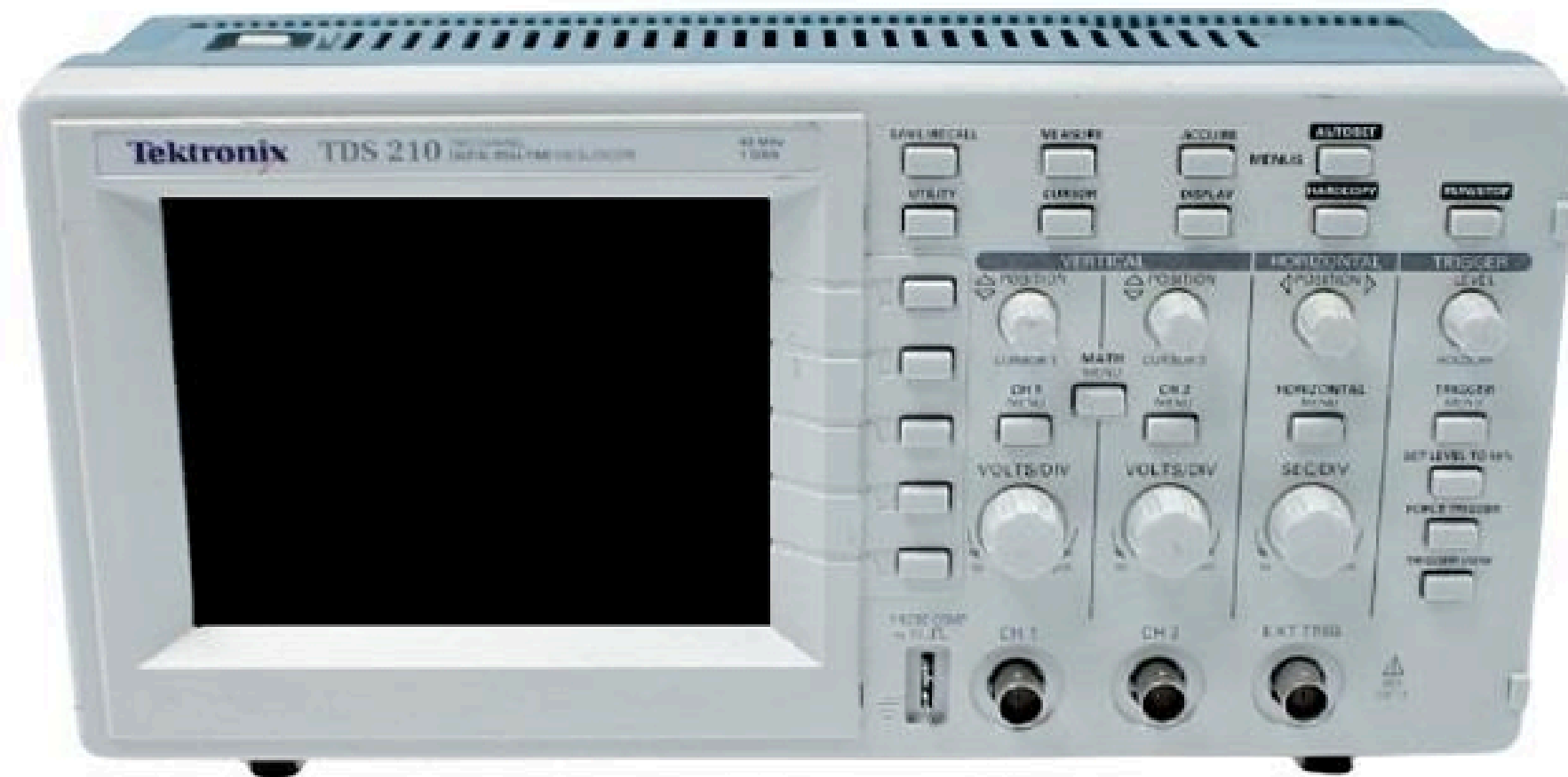
Evaluate the amplitude of the signals using a multimeter and an oscilloscope

Explain how to evaluate uncertainty

Compare different measurement methods

What are the instruments?

Oscilloscope: Tektronix TDS 210



Bandwidth:

- 60 MHz (TDS 210)
- Selectable 20 MHz bandwidth limit
- Sample Rate: 1 GS/s
- Record Length: 2,500 points for each channel

What are the instruments?

Multimeter: 34401A Digital



Capability:

- Measurement:

- 6½ digit resolution
- 10 measurement functions
- Basic accuracy: 0.0035% DC, 0.06% AC
- 1000 V max voltage input, 3 A max current input

- System

- 1000 readings/second
- 512 reading memory

What are the instruments?

Function Generator: Keysight 33220A

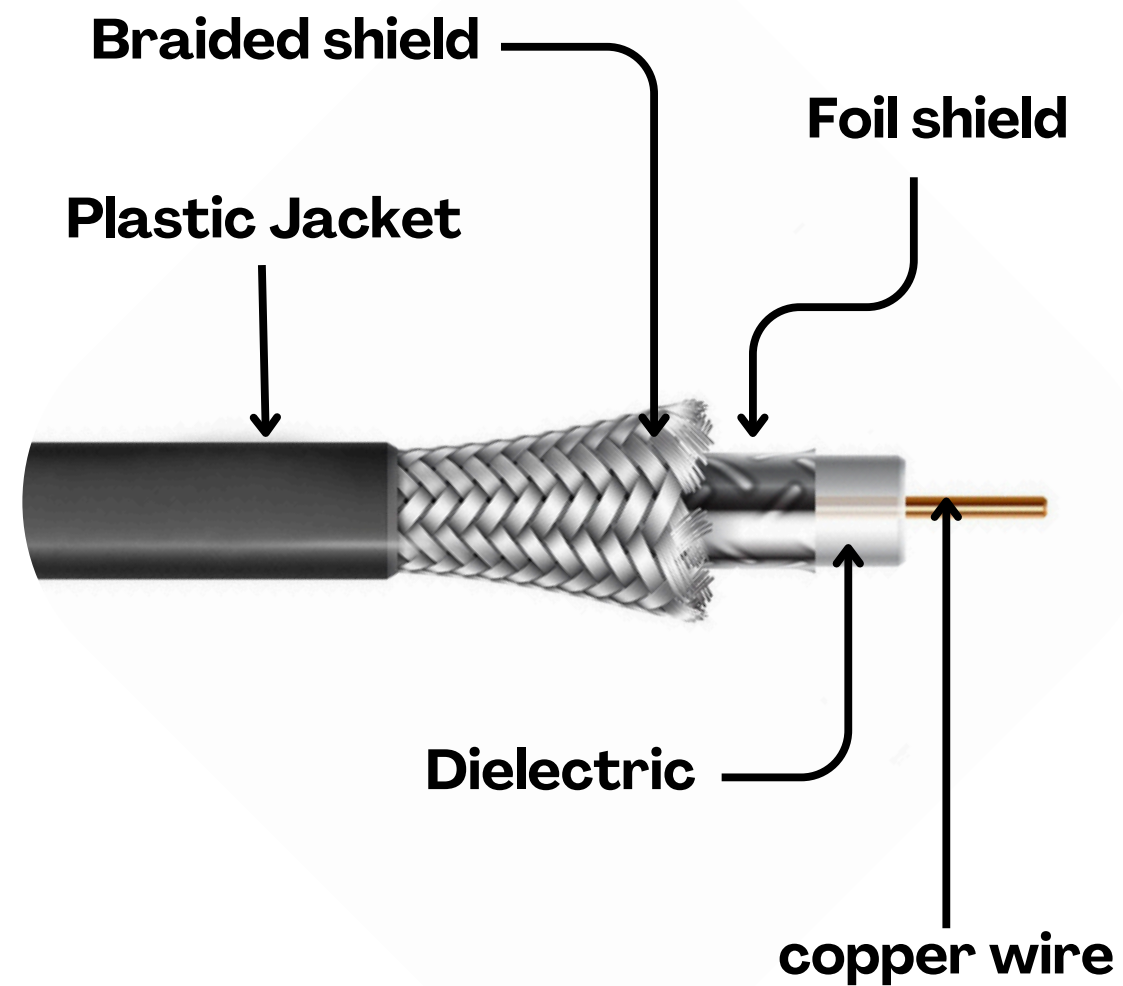


Key Features:

- 20 MHz Sine and Square waveforms
- Pulse, Ramp, Triangle, Noise, and DC waveforms
- 10 mVpp to 10 Vpp amplitude range
- AM, FM, PM, FSK, and PWM modulation types
- 14-bit, 50 MSa/s, 64 k-point arbitrary waveforms

How to link them?

In parallel with a Coaxial Cable



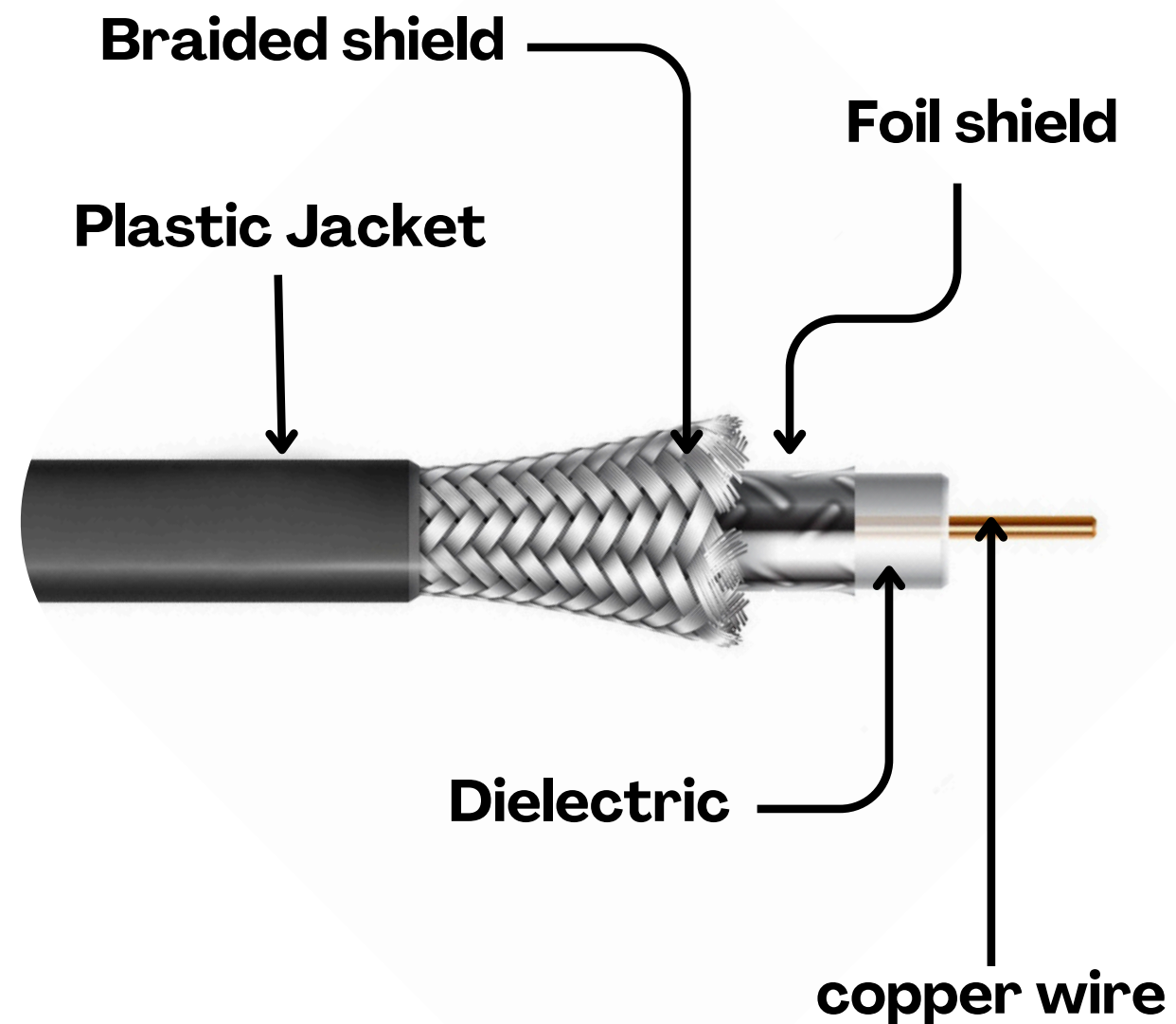
Why in parallel?

- Instruments share the same Voltage

Why a Coaxial Cable?

- Impedance closely matches the output impedance of the connected instruments
- Shield from external noise

How to link them?



Why the Plastic jacket?

- Shield from external interferences

Why the Braided shield?

- Prevent that the cable act as an antenna
- Protecting the signal from external noise
- Acting as a reference ground for the instruments.

Reading
taken
from the
Multimeter

Sinusoidal Signal

f

4 kHz

14 kHz

A

2 V_{pp}

1 V_{pp}

Reading
taken
from the
Multimeter

Square Signal

f

4 kHz

14 kHz

A

2 V_{pp}

1 V_{pp}

Uncertainty:

TypeA

$$U = \sqrt{\frac{1}{(n-1) * n} \sum_{i=1}^n (x_i - \bar{x})^2}$$

Direct measurements that
can be repeated

TypeB

$$U = a \% \text{ Reading} + b \% \text{ Full Scale}$$

Direct measurement, but
we are not able to repeat
the measurement several
times

Type B

What do a and b depends on?

- Last calibration
- Working interval temperature

if T is out of the range:

$$newA = a + (a' * \Delta T)$$

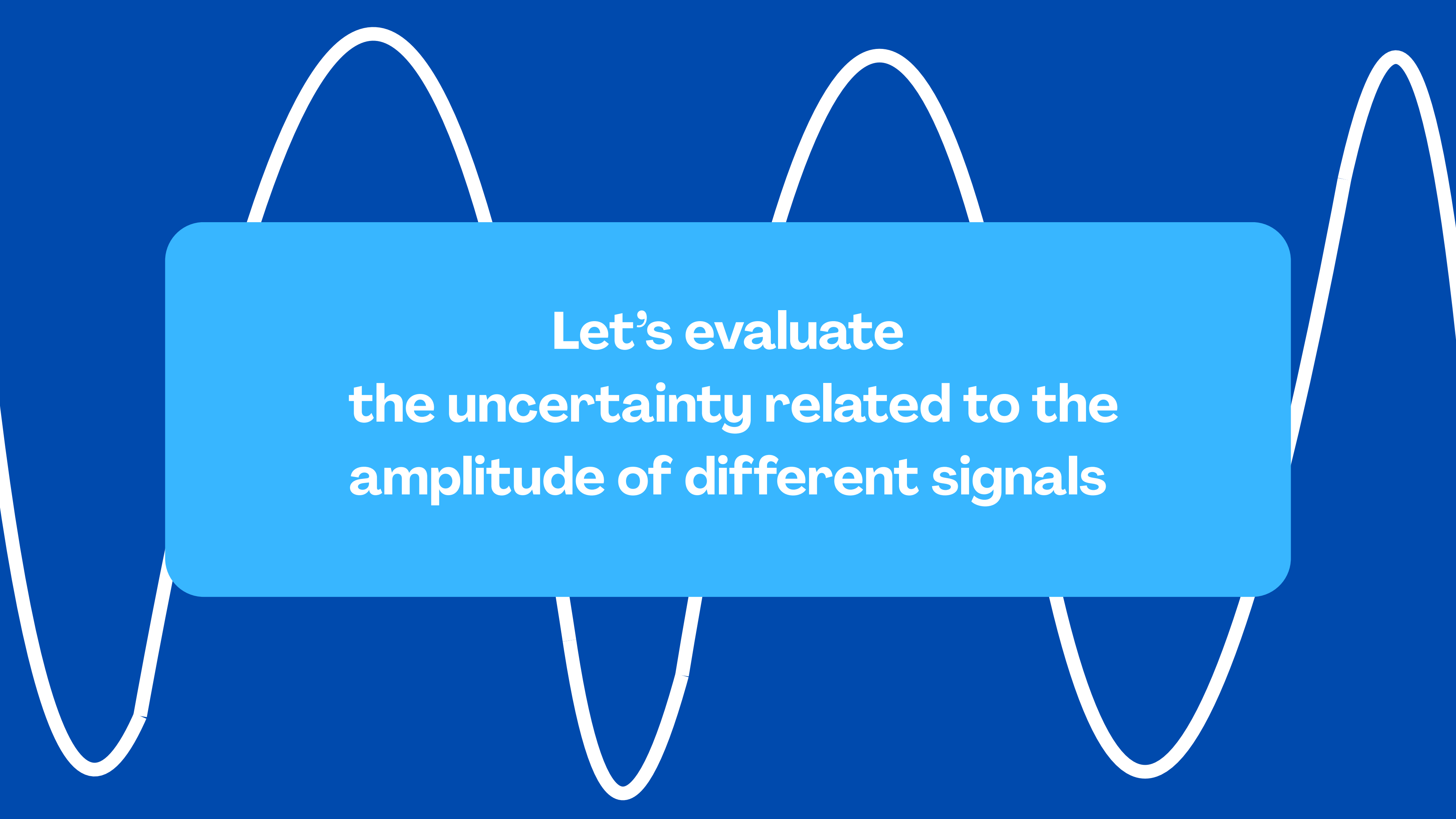
$$newB = b + (b' * \Delta T)$$

$$\Delta T = T - range$$

- Frequency and Amplitude

We take the values from the table:

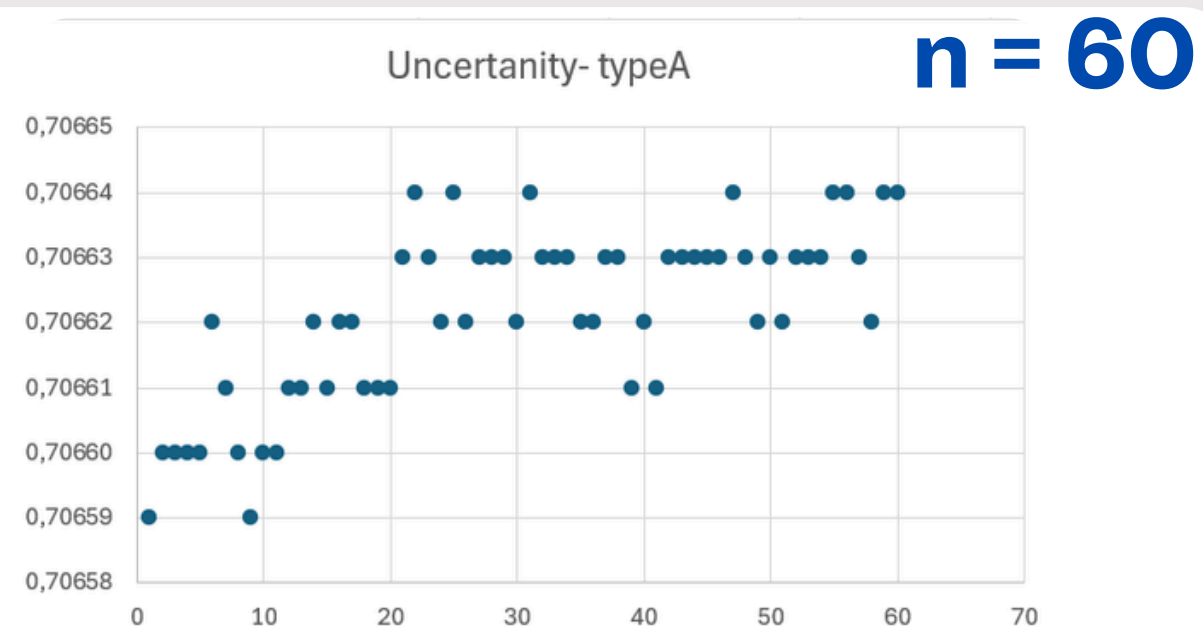
Function	Range ³	Frequency, etc.	24 hour ² 23 ± 1 °C	90 day 23 ± 5 °C	1 year 23 ± 5 °C	Temperature coefficient 0 – 18 °C 28 – 55 °C
DC voltage	100.0000 mV		0.0030 + 0.0030	0.0040 + 0.0035	0.0050 + 0.0035	0.0005 + 0.0005
	1.000000 V		0.0020 + 0.0006	0.0030 + 0.0007	0.0040 + 0.0007	0.0005 + 0.0001
	10.00000 V		0.0015 + 0.0004	0.0020 + 0.0005	0.0035 + 0.0005	0.0005 + 0.0001
	100.0000 V		0.0020 + 0.0006	0.0035 + 0.0006	0.0045 + 0.0006	0.0005 + 0.0001
	1000.000 V		0.0020 + 0.0006	0.0035 + 0.0010	0.0045 + 0.0010	0.0005 + 0.0001
True rms AC voltage ⁴	100.0000 mV	3 – 5 Hz	1.00 + 0.03	1.00 + 0.04	1.00 + 0.04	0.100 + 0.004
		5 – 10 Hz	0.35 + 0.03	0.35 + 0.04	0.35 + 0.04	0.035 + 0.004
		10 Hz – 20 kHz	0.04 + 0.03	0.05 + 0.04	0.06 + 0.04	0.005 + 0.004
		20 – 50 kHz	0.10 + 0.05	0.11 + 0.05	0.12 + 0.05	0.011 + 0.005
		50 – 100 kHz	0.55 + 0.08	0.60 + 0.08	0.60 + 0.08	0.060 + 0.008
		100 – 300 kHz ⁶	4.00 + 0.50	4.00 + 0.50	4.00 + 0.50	0.20 + 0.02
	1.000000 V to 750.000 V	3 – 5 Hz	1.00 + 0.02	1.00 + 0.03	1.00 + 0.03	0.100 + 0.003
		5 – 10 Hz	0.35 + 0.02	0.35 + 0.03	0.35 + 0.03	0.035 + 0.003
		10 Hz – 20 kHz	0.04 + 0.02	0.05 + 0.03	0.06 + 0.03	0.005 + 0.003
		20 – 50 kHz	0.10 + 0.04	0.11 + 0.05	0.12 + 0.04	0.011 + 0.005
		50 – 100 kHz ⁵	0.55 + 0.08	0.60 + 0.08	0.60 + 0.08	0.060 + 0.008
		100 – 300 kHz ⁶	4.00 + 0.50	4.00 + 0.50	4.00 + 0.50	0.20 + 0.02



**Let's evaluate
the uncertainty related to the
amplitude of different signals**

Uncertainty: 4 kHz Sin

TypeA



$$U_A = 1,74 \mu\text{V}$$

TypeB

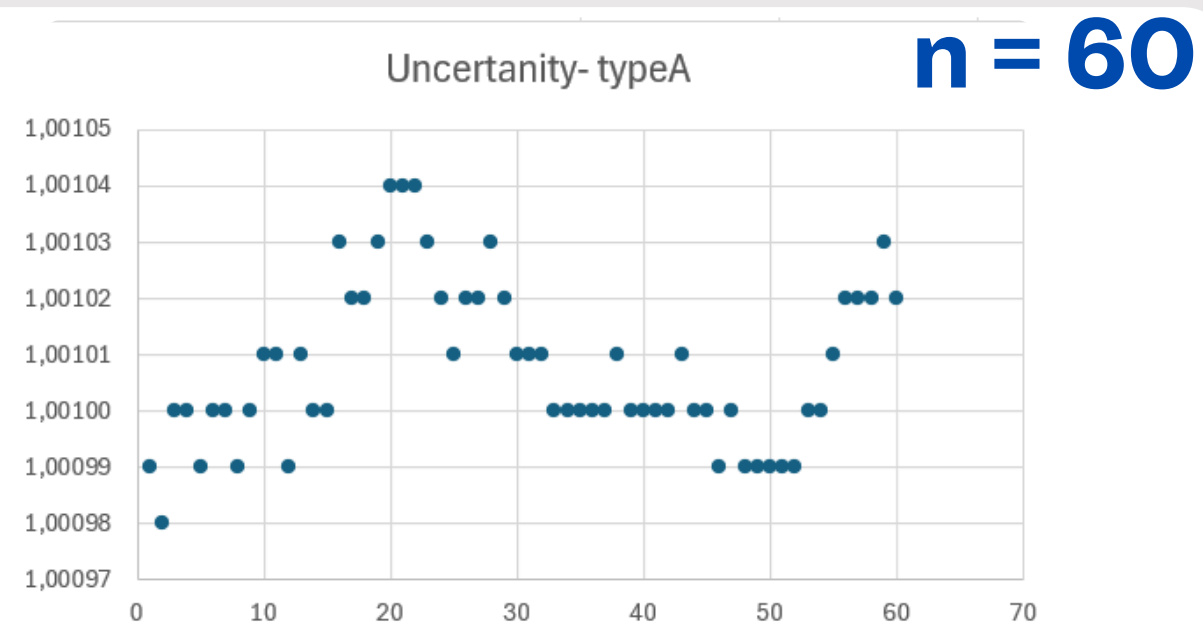
$$a = 0,06\% \quad b = 0,03\%$$

$$\text{FS} = 2 \text{ V} \quad x = 0,70 \text{ V}$$

$$U_B = 1,02 \text{ mV}$$

Uncertainty: 4 kHz Square

TypeA



$$U_A = 1,87 \mu V$$

TypeB

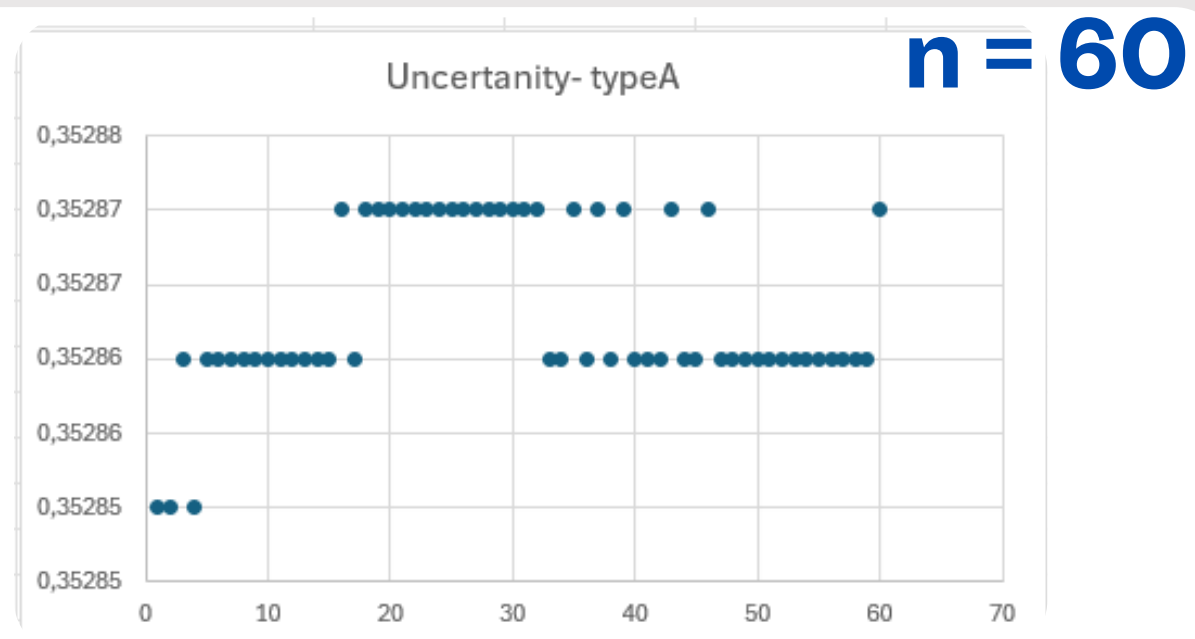
$$a = 0,06\% \quad b = 0,03\%$$

$$FS = 2 V \quad x = 1 V$$

$$U_B = 1,2 mV$$

Uncertainty: 14 kHz Sin

TypeA



$$U_A = 7,3 * 10^{-1} \mu V$$

TypeB

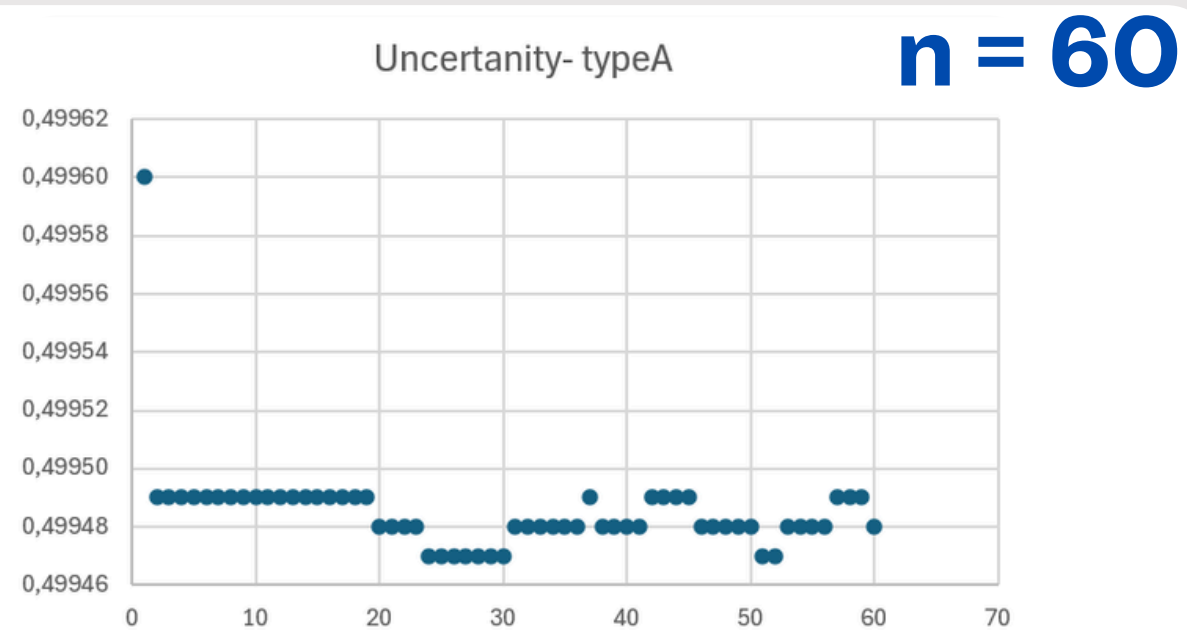
$$a = 0,06\% \quad b = 0,03\%$$

$$FS = 1 V \quad x = 0,35 V$$

$$U_B = 5,12 * 10^{-1} mV$$

Uncertainty: 14 kHz Square

TypeA



$$U_A = 2,16 \mu V$$

TypeB

$$a = 0,06\% \quad b = 0,03\%$$

$$FS = 1 V \quad x = 0,5 V$$

$$U_B = 6 * 10^{-1} mV$$

Results

RMS

The Root Mean Square (RMS) is a crucial statistical tool used to quantify the magnitude of variations, uncertainties, or errors in measurements

$$f_{RMS} = \lim_{T \rightarrow \infty} \sqrt{\frac{1}{2T} \int_{-T}^T [f(t)]^2 dt}$$

Results

Comparison of the signals

Characteristic	Sinusoidal Signal	Rectangular Signal
Waveform	Smooth and continuous, with no abrupt transitions.	Has abrupt transitions between high and low levels.
Harmonic Content	Contains a single frequency (the fundamental).	Contains multiple harmonics (integer multiples of the fundamental frequency).
Frequency Analysis	Easy to analyze in the frequency domain (single peak in the spectrum).	Requires harmonic analysis (multiple peaks in the spectrum).
Typical Applications	Audio, communications, analog circuit testing.	Digital systems, PWM control, clocks, data transmission.
Behavior in Circuits	Predictable response in linear circuits (filters, amplifiers).	May cause transients and require special considerations in digital circuits.

Results

Key concepts that we have learned:

- **Voltage Amplitude Discrepancies**
- **RMS Value Measurement**
- **Waveform Visualization**

Results

Hard Skills Acquired

- **Instrument Operation**
- **Signal Measurement**
- **Impedance Matching**

Results

Hard Skills Acquired

- **Data Analysis and Interpretation**
- **RMS Calculation**
- **Oscilloscope usage**

Results

Soft Skills Developed

- **Experiment Organization**
- **Teamwork and Collaboration**
- **Communication**

Results

Soft Skills Developed

- **Problem-Solving**
- **Critical Thinking**
- **Time Management**
- **Attention to details**

Thank you for your attention

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